1

(a) $P = \underbrace{\frac{RT}{V-b}}_{\text{反発力}} - \underbrace{\frac{a}{V^2}}_{\text{van der Waals 力}}$

$$\left(\frac{\partial P}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial T}{\partial P}\right)_V = -1$$

$$dP = \left(\frac{\partial P}{\partial T}\right)_V dT + \left(\frac{\partial P}{\partial V}\right)_T dV = \frac{\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P}{-\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T} dT - \frac{1}{-\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T} \cdot \frac{dV}{V} = \underbrace{\frac{\alpha}{K}}\, dT - \underbrace{\frac{1}{K} \cdot \frac{dV}{V}}$$

$$dP = \underbrace{\frac{R}{V-b}}\, dT + \underbrace{\left(-\frac{RT}{(V-b)^2} + \frac{2a}{V^3}\right)}\, dV$$

(b) $$\left(-\frac{1}{VK}\right) \cdot (V\alpha) \cdot \frac{1}{\left(\frac{\partial P}{\partial T}\right)_V} = -1 \quad \rightarrow \quad \left(\frac{\partial P}{\partial T}\right)_V = \frac{\alpha}{K}$$

(c) $$\frac{dV}{V} = \alpha(p)\, dT - K(T)\, dp$$

$$\begin{cases} \frac{\partial^2 V}{\partial T \partial P} = V \frac{\partial \alpha}{\partial P} = -A\gamma \\ \frac{\partial^2 V}{\partial P \partial T} = -V \frac{\partial K}{\partial T} = -k\beta \end{cases}$$

$$\rightarrow \quad \ln V = \frac{1}{c}\, \alpha(p) \cdot K(T) \qquad (c = A\gamma = k\beta)$$

$$\therefore \quad c \ln V = A(1-\gamma p) \cdot k(1-\beta(T-T_0))$$

2

$$dS = \frac{1}{T} dU + \frac{P}{T} dV$$

$$\left(\frac{\partial S}{\partial V}\right)_T = \frac{1}{T}\left(\frac{\partial U}{\partial V}\right)_T + \frac{R}{V}$$

$$\left(\frac{\partial S}{\partial T}\right)_V = \frac{1}{T}\left(\frac{\partial U}{\partial T}\right)_V$$

dS が完全微分 $\Leftrightarrow$ $\frac{\partial^2 S}{\partial T \partial V} = \frac{\partial^2 S}{\partial V \partial T}$

$$\Leftrightarrow \frac{\partial}{\partial T}\left(\frac{1}{T}\frac{\partial U}{\partial V}\right) = \frac{\partial}{\partial V}\left(\frac{1}{T}\frac{\partial U}{\partial T}\right)$$

$$\therefore \quad -\frac{1}{T^2}\left(\frac{\partial U}{\partial V}\right)_T = 0$$

3

(a) $dS = \frac{1}{T}dU + \frac{P}{T}dV$ が完全微分 $\Leftrightarrow \frac{\partial^2 S}{\partial T \partial V} = \frac{\partial^2 S}{\partial V \partial T}$

$$\left.\begin{aligned} \left(\frac{\partial S}{\partial V}\right)_T &= \frac{1}{T}\left(\frac{\partial U}{\partial V}\right)_T + \frac{P}{T} \\ \left(\frac{\partial S}{\partial T}\right)_V &= \frac{1}{T}\left(\frac{\partial U}{\partial T}\right)_V \end{aligned}\right\} \Rightarrow \frac{\partial}{\partial T}\left(\frac{1}{T}\frac{\partial U}{\partial V}\right) + \frac{\partial}{\partial T}\left(\frac{P}{T}\right) = \frac{\partial}{\partial V}\left(\frac{1}{T}\frac{\partial U}{\partial T}\right)$$

$$\therefore \left(\frac{\partial U}{\partial V}\right)_T = T^2 \frac{\partial}{\partial T}\left(\frac{P}{T}\right)_V$$

$$= T\left(\frac{\partial P}{\partial T}\right)_V - P \quad \text{」}$$

(b) $$\left(\frac{\partial C_V}{\partial V}\right)_T = \frac{\partial^2 U}{\partial V \partial T} = \frac{\partial}{\partial T}\left\{T\left(\frac{\partial P}{\partial T}\right)_V - P\right\} = T\left(\frac{\partial^2 P}{\partial T^2}\right)_V = 0$$

2つの状態量で決定されるから $C_V$ は $T$ のみ依存 //

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV = C_V dT + \left\{T\left(\frac{\partial P}{\partial T}\right)_V - P\right\} dV$$

$$= C_V dT + \frac{a}{V^2} dV$$

$$\therefore U = \int^T C_V dT - \frac{a}{V} + U_0$$

$$dS = \frac{1}{T}dU + \frac{P}{T}dV = \frac{C_V}{T}dT + \frac{R}{V-b}dV$$

$$\therefore S = \int^T \frac{C_V}{T} dT + R \ln(V-b) + S_0$$

4　張力 $f = aT\left(\frac{l}{l_0} - \left(\frac{l_0}{l}\right)^2\right)$

(a) $dF = -SdT + fdl$　$-S = \left(\frac{\partial F}{\partial T}\right)_l$　$f = \left(\frac{\partial F}{\partial l}\right)_T$ → $F = \int f dl$

$U = F + TS = \int f dl - T\frac{\partial}{\partial T}\int f dl = 0$　$\therefore \left(\frac{\partial U}{\partial l}\right)_T = 0$

(b) $dU = TdS + fdl = 0$　$f = -T\left(\frac{\partial S}{\partial l}\right)_T$

(c) $d'Q = TdS = T\left(\frac{\partial S}{\partial l}\right)_T dl = -fdl$　$(dT = 0)$

$Q = \int_{l_0}^{l} -f dl = aT\left(\frac{3}{2}l_0 - \frac{l^2}{2l_0} - \frac{l_0^2}{l}\right)$

(d) $dS = 0$ → $dT = \left(\frac{\partial T}{\partial l}\right)_S dl = \frac{f}{C_l} dl$　$\left(\frac{\partial S}{\partial T}\right)_l = \frac{C_l}{T}$

$\ln\frac{T}{T_0} = \frac{a}{C_l}\left(\frac{l^2}{2l_0} + \frac{l_0^2}{l} - \frac{3}{2}l_0\right)$

5

## 11

n次元球の体積 $\dfrac{\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2}+1)} r^n$

$$\Omega(E) = \int_{\frac{1}{2m}\sum p_i^2 \le E} dp\, dq = V^N \frac{\pi^{\frac{3}{2}N}}{\Gamma(\frac{3}{2}N+1)} \sqrt{2mE}^{3N} = V^N \frac{(2\pi m E)^{\frac{3}{2}N}}{(\frac{3N}{2})!}$$

$$S(E) = \ln \frac{\Omega(E)}{N!} = \ln \frac{V^N}{(\frac{N}{e})^N} \cdot \frac{(2\pi m E)^{\frac{3}{2}N}}{(\frac{3N}{2e})^{\frac{3}{2}N}} = N \ln \underset{\downarrow}{\frac{V}{N}} \cdot \Big( \frac{4\pi}{3} m \underset{\downarrow}{\frac{E}{N}} \Big)^{\frac{3}{2}} e^{\frac{5}{2}} \propto N$$

v ε

$$\tilde{S}(E) = \ln \Omega(E) = S(E) + N \ln \frac{N}{e}$$ ～～ 示量性に反する

## 12

$$\vec{k} = \frac{2\pi}{L}(n_x, n_y, n_z) \qquad E = \frac{(\hbar \vec{k})^2}{2m} = \frac{h^2}{2mL^2}(n_x^2 + n_y^2 + n_z^2)$$

$$\int g_L(E)\, dE = \frac{4\pi}{3} \sqrt{\frac{2mL^2 E}{h^2}}^3 \rightarrow g_L(E)\, dE = 2\pi \frac{V}{h^3} (2m)^{\frac{3}{2}} \sqrt{E}\, dE$$

古典的 $\displaystyle\int_{E < \frac{p^2}{2m} < E+dE} d^3p\, d^3q = V \cdot \frac{d}{dE}\Big( \frac{4\pi}{3} (2mE)^{\frac{3}{2}} \Big) dE = 2\pi V (2m)^{\frac{3}{2}} \sqrt{E}\, dE$

## 13

(1) $$\langle E \rangle = \frac{\int E e^{-\frac{E}{kT}} dE}{\int e^{-\frac{E}{kT}} dE} = \frac{\Big[ -kT(E+kT) e^{-\frac{E}{kT}} \Big]_0^\infty}{\Big[ -kT e^{-\frac{E}{kT}} \Big]_0^\infty} = kT$$

(2) $$\langle E \rangle = \frac{\sum n\varepsilon e^{-\frac{n\varepsilon}{kT}}}{\sum e^{-\frac{n\varepsilon}{kT}}} = \varepsilon \frac{-\frac{d}{d(\frac{\varepsilon}{kT})} \Big( \frac{1}{1-e^{-\frac{\varepsilon}{kT}}} \Big)}{\frac{1}{1-e^{-\frac{\varepsilon}{kT}}}} = \frac{\varepsilon}{e^{\frac{\varepsilon}{kT}} - 1} \rightarrow kT \qquad (\varepsilon \ll kT)$$

14

(1) $P(n) = {}_N C_n \lambda^n (1-\lambda)^{N-n} = \frac{N!}{n!(N-n)!} \lambda^n (1-\lambda)^{N-n}$

$$a = \langle n \rangle = \sum_{n=0}^{N} n P(n) = N\lambda \sum_{n=1}^{N} \frac{(N-1)!}{(n-1)!(N-n)!} \lambda^{n-1} (1-\lambda)^{N-n}$$

$$= N\lambda \sum_{n=0}^{N-1} \frac{(N-1)!}{n!(N-1-n)!} \lambda^n (1-\lambda)^{N-1-n} = N\lambda$$

$$\sum_{n=0}^{N} n(n-1) P(n) = N(N-1)\lambda^2 \sum_{n=0}^{N-2} \frac{(N-2)!}{n!(N-2-n)!} \lambda^n (1-\lambda)^{N-2-n} = N(N-1)\lambda^2 = \langle n^2 \rangle - \langle n \rangle$$

$$b = \langle (n - \langle n \rangle)^2 \rangle = \langle n^2 \rangle - \langle n \rangle^2 = N(N-1)\lambda^2 + N\lambda - N^2\lambda^2 = N\lambda(1-\lambda)$$

(2) $$F(x) = \sum_{n=0}^{N} P(n) x^n = \{(1-\lambda) + \lambda x\}^N$$

$$= \exp\{N \ln[1 - \lambda(1-x)]\}$$

$$\simeq \exp\{N(-\lambda(1-x))\} = e^{-a(1-x)}$$

$$P(n) = \frac{1}{n!} \left[\frac{d}{dx}\right]^n F(x)\Big|_{x=0} = \frac{a^n}{n!} e^{-a}$$

15

(1) $$\ln P(n) \approx N \ln N - n \ln n - (N-n) \ln (N-n) + n \ln \lambda + (N-n) \ln (1-\lambda)$$

$$(\ln P(n))' = -\ln \frac{n}{N-n} + \ln \frac{\lambda}{1-\lambda} \to 0 \quad (n \to a)$$

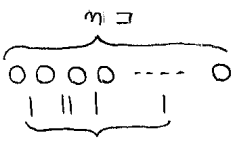
$$(\ln P(n))'' = -\frac{1}{n} - \frac{1}{N-n} \longrightarrow \frac{1}{b} \quad (n \to a)$$

$$\ln P(n) = \underset{0}{\ln P(a)} - \frac{1}{2b}(n-a)^2 + \cdots + O(N^{1-m})(n-a)^m + \cdots \quad (m \geq 3)$$

$$= -\frac{x^2}{2} + x^m O(N^{1-\frac{m}{2}}) \qquad x = \frac{n-a}{\sqrt{b}}$$

## 16

(i) $E_i = s_i \hbar\omega \qquad s_1 + s_2 + \cdots + s_N = n$




$$g(N,n) = {}_{n+N-1}C_{N-1} = \frac{(N+n-1)!}{n!\,(N-1)!}$$

(ii) $S = k \ln g(N,n) = k\left[(N+n)\ln(N+n) - n\ln n - N\ln N\right]$

$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{1}{\hbar\omega}\frac{\partial S}{\partial n} = \frac{k}{\hbar\omega}\ln\frac{N+n}{n} \qquad \frac{N+n}{n} = e^{\frac{\hbar\omega}{kT}} \qquad n = \frac{N}{e^{\frac{\hbar\omega}{kT}} - 1}$$

$$E = n\hbar\omega = \frac{N\hbar\omega}{e^{\frac{\hbar\omega}{kT}} - 1} = N\,\frac{\sum_{p=0}^{\infty} p\hbar\omega\, e^{-p\frac{\hbar\omega}{kT}}}{\sum_{p=0}^{\infty} e^{-p\frac{\hbar\omega}{kT}}}$$

## 17

(i) $$\langle \varepsilon_i \rangle = \frac{\int \varepsilon_i e^{-\sum_j \beta\varepsilon_j} e^{-\beta E'} dq\,dp}{\int e^{-\sum_j \beta\varepsilon_j} e^{-\beta E'} dq\,dp} = \frac{\int \varepsilon_i e^{-\beta\varepsilon_i} dp_i \prod_{j\neq i}\int e^{-\beta\varepsilon_j} dp_j}{\prod_j \int e^{-\beta\varepsilon_j} dp_j} = \frac{\int \varepsilon_i e^{-\beta\varepsilon_i} dp_i}{\int e^{-\beta\varepsilon_i} dp_i}$$

$$= -\frac{\partial}{\partial\beta}\left(\ln\int e^{-\beta\varepsilon_i} dp_i\right)$$

(ii) $$\langle \varepsilon_i \rangle = -\frac{\partial}{\partial\beta}\left(\ln\int e^{-\beta b_i p_i^2} dp_i\right) = -\frac{\partial}{\partial\beta}\ln\sqrt{\frac{\pi}{b_i\beta}} = \frac{1}{2\beta} = \frac{1}{2}kT$$

(iii) $$\sum_{i\geq j} b_{ij} p_i p_j = \sum_{i,j} c_{ij} p_i p_j \qquad (c_{ij} = c_{ji} = \tfrac{1}{2} b_{ij}\ (i\neq j)\quad c_{ii} = b_{ii})$$

$c_{ij}$ 対称行列 ${}^tT C T = \begin{pmatrix} \alpha_1 & & \\ & \ddots & \\ & & \alpha_f \end{pmatrix}$ $\begin{pmatrix} P_1 \\ \vdots \end{pmatrix} = {}^tT \begin{pmatrix} p_1 \\ \vdots \end{pmatrix}$ $\begin{pmatrix} Q_1 \\ \vdots \end{pmatrix} = {}^tT \begin{pmatrix} q_1 \\ \vdots \end{pmatrix}$

$$\sum_i P_i \dot{Q}_i = (P_1 \cdots P_f)\begin{pmatrix} \dot{Q}_1 \\ \vdots \end{pmatrix} = (P_1 \cdots P_f)\,T\cdot {}^tT \begin{pmatrix} \dot{q}_1 \\ \vdots \end{pmatrix} = \sum_i p_i \dot{q}_i \quad \longrightarrow \quad H(q,p) = H(Q,P)$$

$$\dot{Q}_i = t_{ij}\dot{q}_j = t_{ij}\frac{\partial H}{\partial p_j} = \frac{\partial H}{\partial P_j}, \qquad \dot{P}_i = \cdots = -\frac{\partial H}{\partial Q_i}$$

P, Q は正準変数

$$E = \sum \alpha_i P_i^2 + E'(Q_1, \cdots Q_f)$$

$\alpha_i P_i^2$ について等分配則

$$\left\langle \sum_{i\geq j} b_{ij} p^i p^j \right\rangle = \frac{1}{Z}\int dp\,dq \sum_{i\geq j} b_{ij} P_i P_j\, e^{-\frac{1}{kT}\left(\sum_{i\geq j} b_{ij} P_i P_j + E'(q,\cdots)\right)}$$

$$= \frac{1}{Z}\int dp\,dq\ \frac{1}{2}\sum_i \frac{\partial E}{\partial P_i} P_i\, e^{-\frac{E}{kT}} = \frac{kT}{2}\cdot\frac{1}{Z}\int dp\,dq \sum_i P_i \frac{\partial}{\partial P_i} e^{-\frac{E}{kT}} = \frac{kT}{2} f$$

18

(i) $\ln N! = \sum_{n=1}^{N} \ln n = \int_1^N \ln n \, dn = \left[ n \ln n - n \right]_1^N = N \ln N - N$

(ii) $N! = \Gamma(N+1) = \int_0^\infty \underbrace{x^N e^{-x}}_{F(x)} dx$

$\ln F(x) = N \ln x - x$

$(\ln F(x))' = \frac{N}{x} - 1$

$(\ln F(x))'' = -\frac{N}{x^2}$

$\ln F(x) = \ln F(N) + \frac{1}{2}\left(-\frac{1}{N}\right) x^2 + \cdots$

$N! = \left(\frac{N}{e}\right)^N \int_{-\infty}^{\infty} e^{-\frac{x^2}{2N}} dx = \sqrt{2\pi N} \left(\frac{N}{e}\right)^N$

19

(i) $l$ 番目の状態：$G_l$ コに $N_l$ コが分布 $\rightarrow \frac{G_l^{N_l}}{N_l!}$ $\quad \therefore W = \prod_l \frac{G_l^{N_l}}{N_l!}$

(ii) $0 = \frac{\partial}{\partial N_l}\left( S + \alpha N + \beta E \right) = \frac{\partial}{\partial N_l}\left( \ln W + \alpha N + \beta E \right)$

$= \frac{\partial}{\partial N_l} \sum_l \left( N_l \ln G_l - N_l \ln N_l + N_l + \alpha N_l + \beta \epsilon_l N_l \right)$

$= \sum_l \left( \ln G_l - \ln N_l + \alpha + \beta \epsilon_l \right)$ $\quad \frac{G_l}{N_l} = e^{-(\alpha + \beta \epsilon_l)}$

20

(i) $F(\alpha x_i) = \alpha^n F(x_i)$ $\quad \sum_i \frac{\partial F(\alpha x_i)}{\partial(\alpha x_i)} \frac{\partial(\alpha x_i)}{\partial \alpha} = n \alpha^{n-1} F(x_i)$ $\quad \alpha = 1$ とおく。o.k.

(ii) (a) $T = \sum_i \frac{m_i}{2} v_i^2$ $\quad T(\alpha v_i) = \alpha^2 T(v_i)$ 2次の同次式

$2T = \sum_i v_i \frac{\partial T}{\partial v_i} = \sum_i v_i p_i = \frac{d}{dt} \sum_i p_i r_i - \sum_i \dot{p}_i r_i$

(b) $F(t) = \int_0^t f(t) \, dt$ $\quad |F(t)| < M$

$\bar{f} = \left| \frac{1}{T} \int_{-T}^{T} f(t) \, dt \right| = \left| \frac{F(T) - F(-T)}{T} \right| \leq \frac{|F(T)| + |F(-T)|}{T} \leq \frac{2M}{T} \rightarrow 0$

$f = \frac{d}{dt}(r_i p_i)$ $\quad 2\bar{T} = \bar{f} - \overline{\sum \dot{p}_i r_i} = \overline{\sum r_i \frac{\partial U}{\partial r_i}} = n \bar{U}$

(iii) 調和振動子 $U = \sum k_{ij} (r_i - r_j)^2$ $\quad n = 2$

万有引力 $U = \sum g_{ij} / |r_i - r_j|$ $\quad n = -1$

$C = \frac{dE}{d\theta} = \frac{d}{d\theta}(\bar{T} + \bar{U}) = -\frac{d\bar{T}}{d\theta} = -\frac{1}{2} N k < 0$ $\quad \bar{T} = \frac{1}{2} N k \theta$

21

(1) $F = U - TS$ $\qquad dU = TdS - pdV$

$$\frac{\partial}{\partial T}\left(\frac{F}{T}\right)_V = \frac{\partial}{\partial T}\left(\frac{U}{T} - S\right)_V = -\frac{U}{T^2} + \frac{1}{T}\left(\frac{\partial U}{\partial T}\right)_V - \left(\frac{\partial S}{\partial T}\right)_V \qquad \left(\frac{\partial U}{\partial T}\right)_V = T\left(\frac{\partial S}{\partial T}\right)_V$$

(2) $$U = \frac{\sum E_r e^{-\beta E_r}}{\sum_r e^{-\beta E_r}} = -\frac{\partial}{\partial \beta}(\ln Q_N)$$

(3) $$-\frac{\partial}{\partial \beta}(\ln Q_N) = -T^2 \frac{\partial}{\partial T}\left(\frac{F}{T}\right)_V = \frac{1}{k}\frac{\partial}{\partial \beta}\left(\frac{F}{T}\right) = \frac{\partial}{\partial \beta}(F\beta) \qquad F = -\frac{1}{\beta}\ln Q_N$$

(4) $$S = \frac{U-F}{T} = k\beta\left(-\frac{\partial}{\partial \beta}\ln Q_N + \frac{1}{\beta}\ln Q_N\right) = k \ln Q_N + \frac{kT}{Q_N}\left(\frac{\partial Q_N}{\partial T}\right)_V + \frac{\text{const}}{T} \to 0 \quad \begin{pmatrix} S \to 0 \\ T \to 0 \end{pmatrix}$$

$$P = -\left(\frac{\partial F}{\partial V}\right)_T = \frac{kT}{Q_N}\left(\frac{\partial Q_N}{\partial V}\right)_T$$

$$C_V = T\left(\frac{\partial S}{\partial T}\right)_V = T\frac{\partial}{\partial T}\left(k \ln Q_N + kT\frac{\partial}{\partial T}\ln Q_N\right) = kT\frac{\partial^2}{\partial T^2}(T \ln Q_N)$$

(5) $$-k\sum_r P_r \ln P_r = -k\sum_r \frac{e^{-\beta E_r}}{Q_N} \ln \frac{e^{-\beta E_r}}{Q_N} = \frac{k}{Q_N}\sum_r e^{-\beta E_r}(\beta E_r + \ln Q_N)$$

$$= \frac{k}{Q_N}\left(-\beta\frac{\partial Q_N}{\partial \beta} + Q_N \ln Q_N\right) = k \ln Q_N - k\beta\frac{\partial}{\partial \beta}\ln Q_N = S$$

22

$$Z_1 = \frac{1}{h}\int e^{-\frac{p^2}{2mkT} - \frac{m\omega^2}{2kT}q^2} dp\,dq = \frac{1}{h}\sqrt{2\pi mkT}\sqrt{\frac{2\pi kT}{m\omega^2}} = \frac{2\pi kT}{h\omega} = \frac{1}{\beta\hbar\omega}$$

$$Z_N = (\beta\hbar\omega)^{-N}$$

$$U = -\frac{\partial}{\partial \beta}\ln Z_N = \frac{N}{\beta}$$

$$F = -\frac{1}{\beta}\ln Z_N = \frac{N}{\beta}\ln(\beta\hbar\omega)$$

$$P = -\left(\frac{\partial F}{\partial V}\right)_T = 0$$

$$S = \frac{U-F}{T} = Nk(1 - \ln(\beta\hbar\omega))$$

$$C = \frac{\partial U}{\partial T} = Nk$$

23

$$Z_1 = \sum_n e^{-\beta\hbar\omega(n+\frac{1}{2})} = \frac{e^{-\frac{\beta\hbar\omega}{2}}}{1-e^{-\beta\hbar\omega}} = \left[2\sinh\left(\frac{\beta\hbar\omega}{2}\right)\right]^{-1}$$

$$Z_N = \left[2\sinh\left(\frac{\beta\hbar\omega}{2}\right)\right]^{-N}$$

$$U = -\frac{\partial}{\partial\beta}\ln Z_N = N\frac{\partial}{\partial\beta}\ln\left[2\sinh\left(\frac{\beta\hbar\omega}{2}\right)\right] = \frac{N}{2}\hbar\omega \Big/ \tanh\left(\frac{\beta\hbar\omega}{2}\right)$$

$$F = -\frac{1}{\beta}\ln Z_N = \frac{N}{\beta}\ln\left[2\sinh\left(\frac{\beta\hbar\omega}{2}\right)\right]$$

$$P = -\left(\frac{\partial F}{\partial V}\right)_T = 0$$

$$S = \frac{U-F}{T} = Nk\left(\frac{\frac{\beta\hbar\omega}{2}}{\tanh\left(\frac{\beta\hbar\omega}{2}\right)} - \ln\left[2\sinh\left(\frac{\beta\hbar\omega}{2}\right)\right]\right)$$

$$C = \frac{\partial U}{\partial T} = Nk\left(\frac{\frac{\beta\hbar\omega}{2}}{\sinh\left(\frac{\beta\hbar\omega}{2}\right)}\right)^2$$

24

(1) $\sum_r P_r = 1$, $S = -k\sum_r P_r \ln P_r$ $\quad \delta S = \sum_r (-k\ln P_r - k + \alpha)\delta P_r = 0$

$P_r^* = e^{-(1+\frac{\alpha}{k})}$ $\quad \delta^2 S = -\sum_r \frac{k}{P_r}\cdot\delta^2 P_r < 0$ 极大

$\sum_r P_r^* = \Omega P_r^*$ $\quad \therefore P_r^* = \frac{1}{\Omega}$

(2) $\sum_r P_r = 1$, $\sum_r P_r\varepsilon_r = E$ $\quad \delta S = \sum_r (-k\ln P_r - k + \alpha + \beta\varepsilon_r)\delta P_r = 0$

$P_r^* = e^{-(1+\frac{\alpha}{k})} e^{\frac{\beta}{k}\varepsilon_r}$

(3) $\sum_r P_r = 1$, $\sum_r P_r\varepsilon_r = E$, $\sum_r P_r n_r = N$ $\quad \delta S = \sum_r (-k\ln P_r - k + \alpha + \beta\varepsilon_r + \gamma n_r)\delta P_r = 0$

$P_r^* = e^{-(1-\frac{\alpha}{k})} e^{\frac{\beta}{k}\varepsilon_r + \frac{\gamma}{k}n_r}$

## 25

(1) $S = -k \sum_{r,s} P_{r,s} \ln P_{r,s} = -k \sum_{r,s} P_r^{(1)} P_s^{(2)} (\ln P_r^{(1)} + \ln P_s^{(2)})$

$= -k \sum_r P_r^{(1)} \ln P_r^{(1)} \underbrace{\sum_s P_s^{(2)}}_{1} - k \underbrace{\sum_r P_r^{(1)}}_{1} \sum_s P_s^{(2)} \ln P_s^{(2)} = S_1 + S_2$

(2) $S_1 + S_2 - S = -k \sum_r P_r^{(1)} \ln P_r^{(1)} - k \sum_s P_s^{(2)} \ln P_s^{(2)} + k \sum_{rs} P_{rs} \ln P_{rs}$

$= -k \sum_r (\sum_s P_{rs}) \ln P_r^{(1)} - k \sum_s (\sum_r P_{rs}) \ln P_s^{(2)} + k \sum_{rs} P_{rs} \ln P_{rs}$

$= -k \sum_{rs} (P_{rs} \ln P_r^{(1)} + P_{rs} \ln P_s^{(2)} - P_{rs} \ln P_{rs})$

$= -k \sum_{rs} P_{rs} \ln \left( \frac{P_r^{(1)} P_s^{(2)}}{P_{rs}} \right) \geq -k \sum_{rs} P_{rs} \left( \frac{P_r^{(1)} P_s^{(2)}}{P_{rs}} - 1 \right)$

$= -k \sum_{rs} (P_r^{(1)} P_s^{(2)} - P_{rs}) = 0$

## 26

$P_0 \propto e^{-\beta E_0}$

$P_1 \propto 3 e^{-\beta E_1}$

$\begin{cases} e^{-\beta E_0} = 3 e^{-\beta E_1} \longrightarrow \beta (E_1 - E_0) = \ln 3 \\ E_1 - E_0 = 1500\,^\circ K \end{cases}$

$T = \frac{1500}{\ln 3} = 1365\,^\circ K$

## 27

(1) $H = \sum_{i=1}^{3N} \frac{p_i^2}{2m}$ $\quad Z = \frac{1}{h^{3N} N!} \int e^{-\beta H} d\Gamma$ $\quad d\Gamma = \prod_{i=1}^{3N} dq_i dp_i$

$= \frac{V^N}{h^{3N} N!} (2\pi m k T)^{\frac{3N}{2}}$

$F = -kT \ln Z = -NkT \left\{ 1 + \frac{3}{2} \ln T + \ln \frac{V}{N} + \ln \frac{(2\pi m k)^{\frac{3}{2}}}{h^3} \right\}$

$S = -\left(\frac{\partial F}{\partial T}\right)_V = Nk \left\{ \frac{5}{2} + \frac{3}{2} \ln T + \ln \frac{V}{N} + \ln \frac{(2\pi m k)^{\frac{3}{2}}}{h^3} \right\}$

(2) $mT = \text{const}$

$300K \cdot 40 = T_{He} \cdot 4 \qquad T_{He} = 3000K$

28

$F = U - TS$

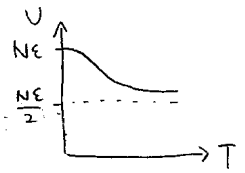
$dF = -pdV - SdT$

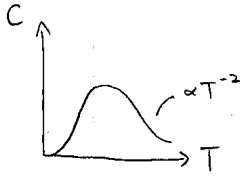
(1) $Z = \left(1 + e^{-\frac{\varepsilon}{kT}}\right)^N \qquad F = -NkT \ln\left(1 + e^{-\frac{\varepsilon}{kT}}\right)$

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V} = -kT \ln\left(1 + e^{-\frac{\varepsilon}{kT}}\right)$$

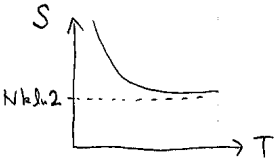
(2) $U = -\frac{\partial}{\partial \beta} \ln Z = N\varepsilon \Big/ 1 + e^{-\frac{\varepsilon}{kT}}$

$$S = \frac{U - F}{T} = Nk\left(\frac{\frac{\varepsilon}{kT}}{1 + e^{-\frac{\varepsilon}{kT}}} + \ln\left(1 + e^{-\frac{\varepsilon}{kT}}\right)\right)$$




(3) $C = \frac{\partial U}{\partial T} = Nk \frac{\left(\frac{\varepsilon}{kT}\right)^2 e^{-\frac{\varepsilon}{kT}}}{\left(1 + e^{-\frac{\varepsilon}{kT}}\right)^2} = Nk\left(\frac{\frac{\varepsilon}{2kT}}{\cosh\left(\frac{\varepsilon}{2kT}\right)}\right)^2$




29

$$\left.\begin{aligned} U &= \tfrac{3}{2}NkT \\ PV &= NkT \end{aligned}\right\} \quad U' = \tfrac{3}{2}PV + 10^4[J] = \tfrac{3}{2}NkT'$$

$1\,atm = 10^5\,N/m^2 = 10^5\,J/m^3$

$$T' = \left(1 + \frac{2}{3PV} \cdot 10^4[J]\right)T = 900\,K$$

30

(1) $2N = N_+ + N_- \qquad 2M = N_+ - N_-$

$$W(L) = \frac{(2N)!}{N_+!\,N_-!} = \frac{(2N)!}{(N+M)!\,(N-M)!} = \frac{(2N)!}{\left(N + \frac{L}{2a}\right)!\left(N - \frac{L}{2a}\right)!}$$

$$S = k \ln W(L) = k\left\{2N \ln 2N - \left(N + \tfrac{L}{2a}\right)\ln\left(N + \tfrac{L}{2a}\right) - \left(N - \tfrac{L}{2a}\right)\ln\left(N - \tfrac{L}{2a}\right)\right\}$$

(2) $F = -kT \ln W(L)$

$S = -\left(\frac{\partial F}{\partial T}\right)_V = k \ln W(L)$

(3) $P = \left.\frac{\partial F}{\partial L}\right|_{T,V} = \frac{kT}{2a} \ln \frac{N + \frac{L}{2a}}{N - \frac{L}{2a}} = \frac{kT}{2a} \ln\left(1 + \frac{1}{\frac{aN}{L} - \frac{1}{2}}\right)$



## 31

(1) $E = \frac{1}{2}ax^2 - Mgx = \frac{1}{2}a\left(x - \frac{Mg}{a}\right)^2 - \frac{M^2g^2}{2a}$

$$\bar{x} = \frac{\int x e^{-\frac{E}{kT}} dx}{\int e^{-\frac{E}{kT}} dx} = \frac{\int (x' + \frac{Mg}{a}) e^{-\frac{a}{2kT}x'^2} dx'}{\int e^{-\frac{a}{2kT}x'^2} dx'} = \frac{Mg}{a}$$

(2) $$\overline{(x-\bar{x})^2} = \frac{\int (x-\bar{x})^2 e^{-\frac{E}{kT}} dx}{\int e^{-\frac{E}{kT}} dx} = \frac{\int x'^2 e^{-\frac{a}{2kT}x'^2} dx'}{\int e^{-\frac{a}{2kT}x'^2} dx'} = -\frac{\partial}{\partial(\frac{a}{2kT})} \ln \int e^{-\frac{a}{2kT}x'^2} dx'$$

$$= -\frac{\partial}{\partial \alpha} \ln \sqrt{\frac{\pi}{\alpha}} = \frac{1}{2\alpha} = \frac{kT}{a}$$

(3) $\sqrt{\frac{kT}{a}} \ll \frac{Mg}{a}$ $\qquad M \gg \frac{\sqrt{akT}}{g}$

## 32

(1) $$Z_1 = \sum_{j=-J}^{J} e^{\beta j \mu H} = e^{-\beta J \mu H} \frac{1 - e^{\beta(2J+1)\mu H}}{1 - e^{\beta \mu H}} = \frac{\sinh\left[(J+\frac{1}{2})\beta\mu H\right]}{\sinh(\frac{1}{2}\beta\mu H)}$$

$$E = -\frac{\partial}{\partial \beta} \ln Z_1^N = -N\mu H\left[(J+\tfrac{1}{2})\coth\left[(J+\tfrac{1}{2})\beta\mu H\right] - \tfrac{1}{2}\coth(\tfrac{1}{2}\beta\mu H)\right] = -M \cdot H$$

$$M = N\mu\left[(J+\tfrac{1}{2})\coth\left[(J+\tfrac{1}{2})\beta\mu H\right] - \tfrac{1}{2}\coth(\tfrac{1}{2}\beta\mu H)\right]$$

$$= \frac{N\mu^2\beta H}{3} J(J+1)$$

$$\coth x = \frac{e^{2x}+1}{e^{2x}-1} = 1 + \frac{2}{e^{2x}-1} = 1 + \frac{1}{x + x^2 + \frac{2}{3}x^3 \cdots}$$

$$= 1 + \frac{1}{x}\left(1 - x + \frac{x^2}{3} \cdots\right)$$

$$= \frac{1}{x} + \frac{x}{3} + \cdots$$

(2) $$\chi = \left.\frac{\partial M}{\partial H}\right|_{H=0} = \frac{1}{3} J(J+1) \frac{N\mu^2}{kT}$$

(3) $$Z_1 = \int e^{\beta J \mu H \cos\theta} \frac{d\Omega}{4\pi} = \int e^{\beta J \mu H \cos\theta} \frac{\sin\theta d\theta d\phi}{4\pi} = -\frac{1}{2}\int_1^{-1} e^{\beta J \mu H \cos\theta} d\cos\theta$$

$$= \frac{\sinh(\beta J \mu H)}{\beta J \mu H}$$

$$E = -\frac{\partial}{\partial \beta} \ln Z_1^N = -N\left[J\mu H \coth(\beta J \mu H) - \frac{1}{\beta}\right]$$

$$= -N\frac{\beta(J\mu H)^2}{3}$$

$$M = \frac{N\beta(J\mu)^2}{3} H$$

$$\chi = \frac{N(\mu J)^2}{3kT}$$

33

$$\overline{E^n} = \int E^n e^{-\beta E}\, dE \,/\, Z = (-1)^n \frac{Z^{(n)}}{Z}$$

$$\frac{\partial \bar{E}}{\partial \beta} = -\frac{\partial}{\partial \beta}\left(\frac{Z'}{Z}\right) = -\frac{Z''}{Z} + \frac{Z'^2}{Z^2} = -\overline{E^2} + \bar{E}^2 = -\overline{(E-\bar{E})^2}$$

$$\frac{\partial^2 \bar{E}}{\partial \beta^2} = -\frac{Z'''}{Z} + 3\frac{Z''Z'}{Z^2} - 2\frac{Z'^3}{Z^3} = \overline{E^3} - 3\overline{E^2}\,\bar{E} + 2\bar{E}^3 = \overline{(E-\bar{E})^3}$$

$$\frac{\partial^3 \bar{E}}{\partial \beta^3} = -\frac{Z''''}{Z} + 4\frac{Z'''Z'}{Z^2} - 12\frac{Z''Z'^2}{Z^3} + 3\frac{Z''^2}{Z^2} + 6\frac{Z'^4}{Z^4} = -\overline{E^4} + 4\overline{E^3}\,\bar{E} - 12\overline{E^2}\bar{E}^2 + 3\overline{E^2} + 6\bar{E}^4$$

$$= -\overline{(E-\bar{E})^4} + 3\left(\frac{\partial \bar{E}}{\partial \beta}\right)^2$$

$$\frac{\partial \bar{E}}{\partial \beta} = -kT^2 \frac{\partial \bar{E}}{\partial T} = -kT^2 C_v$$

$$\frac{\partial^2 \bar{E}}{\partial \beta^2} = k^2 T^2 \frac{\partial}{\partial T}\left(T^2 C_v\right) = k^2\left(T^4 \frac{\partial C_v}{\partial T} + 2T^3 C_v\right)$$

$$\frac{\partial^3 \bar{E}}{\partial \beta^3} = -k^3 T^2 \frac{\partial}{\partial T}\left(T^4 \frac{\partial C_v}{\partial T} + 2T^3 C_v\right) = -k^3\left(T^6 \frac{\partial^2 C_v}{\partial T^2} + 6T^5 \frac{\partial C_v}{\partial T} + 6T^4 C_v\right)$$

$$\bar{E} = \frac{3}{2}NkT \qquad C_v = \frac{3}{2}Nk$$

$$kT^2 C_v = k\left(\frac{\bar{E}}{C_v}\right)^2 C_v = \bar{E}^2 \frac{2}{3N}$$

$$k^2\left(T^4 \frac{\partial C_v}{\partial T} + 2T^3 C_v\right) = 2k^2\left(\frac{\bar{E}}{C_v}\right)^3 C_v = 2\bar{E}^3\left(\frac{2}{3N}\right)^2$$

34

$$\Xi = \sum_{N\tau} e^{\beta(\mu N - \varepsilon N)} \qquad \langle N \rangle = \frac{1}{\beta}\frac{\partial \Xi}{\partial \mu} \,/\, \Xi$$

1. $$\langle N^2 \rangle - \langle N \rangle^2 = \frac{1}{\beta^2}\left(\frac{\Xi''}{\Xi} - \frac{\Xi'^2}{\Xi^2}\right) = \frac{1}{\beta^2}\left(\frac{\Xi'}{\Xi}\right)' = \frac{1}{\beta}\frac{\partial}{\partial \mu}\langle N \rangle$$

2. $$\Xi = \sum \lambda^N Z = \sum \lambda^N \frac{Z_1^N}{N!} = e^{\lambda Z_1} \qquad \lambda = e^{\beta\mu}$$

$$\langle N \rangle = \lambda \frac{\partial}{\partial \lambda} \ln \Xi = \lambda Z_1 \qquad \langle N^2 \rangle - \langle N \rangle^2 = \lambda \frac{\partial}{\partial \lambda}\langle N \rangle = \langle N \rangle$$

11. $$\Xi_\tau = \sum_n e^{\beta(\mu - \varepsilon_\tau)n} = \left(1 \mp e^{\beta(\mu-\varepsilon)}\right)^{\mp} \qquad \begin{matrix} - \text{ Bose} \\ + \text{ Fermi} \end{matrix}$$

$$\langle n_\tau \rangle = \frac{1}{\beta}\frac{\partial}{\partial \mu}\left[\mp \ln\left(1 \mp e^{\beta(\mu-\varepsilon)}\right)\right] = \frac{1}{e^{\beta(\varepsilon-\mu)} \mp 1}$$

$$\langle n_\tau^2 \rangle - \langle n_\tau \rangle^2 = \frac{1}{\beta}\frac{\partial}{\partial \mu}\left(\frac{1}{e^{\beta(\varepsilon-\mu)} \mp 1}\right) = \frac{1}{\beta}\cdot\langle n_\tau \rangle^2 \left(-\beta e^{\beta(\varepsilon-\mu)}\right) = -\langle n \rangle\left(1 \pm \langle n \rangle\right) \quad ?$$

35

$f(t) = \langle e^{tX} \rangle \qquad f^{(n)}(t)\big|_{t=0} = \langle X^n \rangle$

[1] $z_n = \frac{d^n}{dt^n} g(t)\Big|_{t=0} = \frac{d^n}{dt^n} (\log f(t))\Big|_{t=0}$

$z_1 = \frac{f'}{f} = \frac{\langle x \rangle}{\langle 1 \rangle} \qquad z_2 = \frac{f''}{f} - \frac{f'^2}{f^2} = \langle x^2 \rangle - \langle x \rangle^2$

$z_3 = \frac{f'''}{f} - 3\frac{f''f'}{f^2} + 2\frac{f'^3}{f^3} = \langle x^3 \rangle - 3\langle x^2 \rangle \langle x \rangle + 2\langle x \rangle^2$

[2] $f(t) = \langle e^{tx} \rangle = \int e^{tx} \frac{1}{\sqrt{2\pi v}} e^{-\frac{(x-m)^2}{2v}} dx = e^{mt + \frac{v}{2}t^2} \int \frac{1}{\sqrt{2\pi v}} e^{-\frac{(x-(m+vt))^2}{2v}} dx$

$g(t) = \ln f(t) = \underset{z_1}{mt} + \underset{z_2}{\frac{1}{2} v t^2}$ ⌟ $\qquad z_{n \geq 3} = 0$

[3] $f(t) = \sum_{x=0}^{\infty} e^{tx} \frac{a^x e^{-a}}{x!} = e^{-a} \sum_{x=0}^{\infty} \frac{1}{x!} (e^t a)^x = e^{-a + e^t a}$

$g(t) = \ln f(t) = -a + e^t a = -a + a \sum_{n=0}^{\infty} \frac{t^n}{n!} = \sum_{n=1}^{\infty} \frac{t^n}{n!} a \qquad \therefore z_n = a$

36

$z_1 = \frac{1}{h^3} \int e^{-\beta \frac{\vec{p}^2}{2m}} d^3p\, d^3q = \frac{V}{h^3} (2\pi m kT)^{\frac{3}{2}} \qquad Z_N = \frac{z_1^N}{N!} = \left(\frac{eV}{N}\right)^N \left(\frac{2\pi m kT}{h^2}\right)^{\frac{3N}{2}}$

$F = -NkT \left\{ 1 + \ln \frac{V}{N} + \frac{3}{2} \ln \frac{2\pi m kT}{h^2} \right\}$

$\mu = \frac{\partial F}{\partial N} = kT \ln \left[ \left(\frac{N}{V}\right) \left(\frac{h^2}{2\pi m kT}\right)^{\frac{3}{2}} \right]$

37 $\qquad dF = -S\,dT + X\,dL$

$X = AT = \left(\frac{\partial F}{\partial L}\right)_T \qquad F = ATL + f(T)$

$S = -\left(\frac{\partial F}{\partial T}\right)_L = -AL - \left(\frac{\partial f}{\partial T}\right)_L = -AL - f'(T)$

$U = F + TS = f(T) - T f'(T)$

38

$$g(\omega) = \begin{cases} \frac{9N}{\omega_D^3}\omega^2 & (\omega < \omega_D) \\ 0 & (\omega_D < \omega) \end{cases}$$

$$Z = \prod_\omega \left( \sum_n e^{-\beta n\hbar\omega} \right)^{g(\omega)\Delta\omega} = \prod_\omega \left( \frac{1}{1-e^{-\beta\hbar\omega}} \right)^{g(\omega)\Delta\omega}$$

$$\ln Z = -\int_0^{\omega_D} g(\omega)\, d\omega \ln\left(1-e^{-\beta\hbar\omega}\right)$$

$$E = -\frac{\partial}{\partial\beta}\ln Z = \int g(\omega)\frac{\hbar\omega\, e^{-\beta\hbar\omega}}{1-e^{-\beta\hbar\omega}}\, d\omega = \frac{9N}{(\hbar\omega_D)^3}\cdot\frac{1}{\beta^4}\int_0^{\beta\hbar\omega_D}\frac{x^3}{e^x-1}\,dx \qquad x = \beta\hbar\omega$$

$$= \frac{3N}{\beta} D(\beta\hbar\omega_D) \qquad D(y) = \frac{3}{y^3}\int_0^y \frac{x^3}{e^x-1}\,dx$$

$$C_V = \frac{\partial E}{\partial T} = 3Nk\left(D(x_D) - x_D D'(x_D)\right)$$

$$= 3Nk\left(4D(x_D) - \frac{3x_D}{e^{x_D}-1}\right)$$

高温: $C_V = 3Nk$

低温: $C_V = 3Nk\left(\frac{12M}{x_D^3} - 3x_D e^{-x_D}\right)$

$$= 3Nk\left(\frac{kT}{\hbar\omega_D}\right)^3 \frac{4\pi^4}{5}$$

$$\frac{x^3}{e^x-1} = \left(\frac{x^3}{3}D(x)\right)' = x^2 D(x) + \frac{x^3}{3}D'(x)$$

$$D(x) \to \frac{3}{x^3}\int_0^x \frac{x^3}{x}\,dx = 1 \quad (x\to 0)$$

$$D(x) \to \frac{3}{x^3}\int_0^\infty \frac{x^3}{e^x-1}\,dx = \frac{3M}{x^3} \quad (x\to\infty)$$

$$M = \frac{\pi^4}{15}$$

39

(1) $\Omega = -pV = -\frac{1}{\beta}\ln \square \qquad d\Omega = -S\,dT - P\,dV - N\,d\mu$

$$\bar{N} = -\left(\frac{\partial\Omega}{\partial\mu}\right)_{T,V} = \frac{1}{\beta}\frac{\partial}{\partial\mu}\ln\square \quad , \quad \bar{P} = -\left(\frac{\partial\Omega}{\partial V}\right)_{T,\mu} = \frac{1}{\beta}\frac{\partial}{\partial V}\ln\square$$

$$\square = \sum_{n,r} e^{\beta(\mu N - E_r)} \qquad \mu\bar{N} - \bar{E} = \frac{\partial}{\partial\beta}\ln\square \qquad \bar{E} = -\frac{\partial}{\partial\beta}\ln\square + \frac{\mu}{\beta}\frac{\partial}{\partial\mu}\ln\square$$

(2) $$\square = \sum_N \frac{1}{N!\,h^{3N}}\int e^{\beta(\mu N - \sum\frac{p^2}{2m})}\,dp\,dq = \sum_N \frac{V^N}{N!\,h^{3N}} e^{\beta\mu N}\left(\frac{2\pi m}{\beta}\right)^{\frac{3N}{2}} \qquad \lambda_T^2 = \frac{\beta h^2}{2\pi m}$$

$$= \sum_N \frac{V^N}{N!} e^{\beta\mu N}\lambda_T^{-3N} = \sum_N \frac{1}{N!}\left(\frac{Ve^{\beta\mu}}{\lambda_T^3}\right)^N = \exp\left(\frac{Ve^{\beta\mu}}{\lambda_T^3}\right)$$

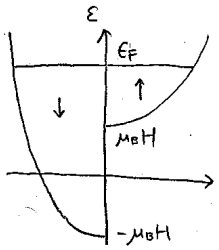
$$\bar{N} = \frac{1}{\beta}\frac{\partial}{\partial\mu}\ln\square = \frac{Ve^{\beta\mu}}{\lambda_T^3} \qquad \bar{P} = \frac{1}{\beta}\frac{\partial}{\partial V}\ln\square = \frac{e^{\beta\mu}}{\beta\lambda_T^3} = \frac{\bar{N}}{\beta V} = \frac{\bar{N}kT}{V}$$

$$\bar{E} = -\frac{\partial}{\partial\beta}\ln\square + \mu\bar{N} = -\mu\frac{Ve^{\beta\mu}}{\lambda_T^3} + \frac{3}{2}\frac{Ve^{\beta\mu}}{\lambda_T^3}\frac{1}{\beta} + \mu\bar{N} = \frac{3\bar{N}}{2\beta} = \frac{3}{2}\bar{N}kT$$

40 → 34 N.

41

$\varepsilon = \frac{p^2}{2m} \pm \mu_B H$




$T = 0°K$ : $N_\pm = \frac{V}{h^3}\frac{4\pi}{3} p_\pm^3$ $\qquad \epsilon_F = \frac{p_\pm^2}{2m} \pm \mu_B H$

$$M = -\mu_B (N_+ - N_-) = -\frac{4\pi V}{3h^3}\mu_B (p_+^3 - p_-^3) = \cdots$$

$$= -\frac{4\pi V}{3h^3}\mu_B (2m)^{\frac{3}{2}} \left\{ (\epsilon_F - \mu_B H)^{3/2} - (\epsilon_F + \mu_B H)^{3/2} \right\} \qquad \epsilon_F \gg \mu_B H$$

$$= \frac{4\pi V}{h^3}\mu_B (2m\mu_0)^{\frac{3}{2}} \frac{\mu_B H}{\epsilon_F} = \frac{3}{2}\frac{\mu_B^2 N}{\epsilon_F} H$$

$$N = N_+ + N_- = \frac{4\pi V}{3h^3}(p_+^3 + p_-^3) = \frac{4\pi V}{3h^3}(2m)^{\frac{3}{2}} 2\epsilon_F^{\frac{3}{2}}$$

$$\chi = \frac{\partial M}{\partial H} = \frac{3}{2}\frac{\mu_B^2 N}{\epsilon_F} = \frac{3}{2}\frac{\mu_B^2 N}{kT_0}$$

状態密度 $D(\varepsilon) = \frac{2\pi V}{h^3}(2m)^{\frac{3}{2}}\varepsilon^{\frac{1}{2}}$ ← スピンを考えない

$$M = \mu_B \left\{ \int \frac{D(\varepsilon)d\varepsilon}{e^{\beta(\varepsilon - \mu_B H - \mu_0)} + 1} - \int \frac{D(\varepsilon)d\varepsilon}{e^{\beta(\varepsilon + \mu_B H - \mu_0)} + 1} \right\}$$

$$= \mu_B \int_0^\infty d\varepsilon \left( D(\varepsilon + \mu_B H) - D(\varepsilon - \mu_B H) \right) f(\varepsilon) \qquad f(\varepsilon) = \frac{1}{e^{\beta(\varepsilon - \mu_0)} + 1}$$

$$= 2\mu_B^2 H \int_0^\infty D'(\varepsilon) f(\varepsilon) d\varepsilon \qquad D(0) = 0$$

$$= 2\mu_B^2 H D(\epsilon_F) \qquad (T = 0°K \text{ . 完全縮退})$$

## 42

(1) $S' = k \ln {}_P C_M = k \ln \frac{P!}{M!(P-M)!} = k[P\ln P - M\ln M - (P-M)\ln(P-M)]$

$F' = U' - TS' = -M\varepsilon - kT[P\ln P - M\ln M - (P-M)\ln(P-M)]$

$\mu' = \frac{\partial F'}{\partial M} = -\varepsilon - kT \ln \frac{P-M}{M}$

(2) $\mu = kT \ln\left[\frac{N}{V}\left(\frac{h^2}{2\pi m kT}\right)^{3/2}\right] = -\varepsilon - kT \ln\left(\frac{1}{\theta} - 1\right)$

$\frac{1-\theta}{\theta} \cdot \frac{Q-\theta P}{V}\left(\frac{h^2}{2\pi m kT}\right)^{3/2} = e^{-\frac{\varepsilon}{kT}} \qquad \alpha = V\left(\frac{2\pi m kT}{h^2}\right)^{3/2} e^{-\frac{\varepsilon}{kT}}$

$(1-\theta)(Q-\theta P) = \alpha\theta$

低温 $\alpha \ll 1$ $\theta = 1$ $(P \ll Q)$ or $\theta = \frac{Q}{P}$ $(P > Q)$

高温 $\alpha \gg 1$ $\theta = \frac{Q}{P+Q+\alpha} = \frac{Q}{Q+\alpha}$

## 43

$E = -\epsilon + \hbar\omega\left(n + \frac{1}{2}\right)$

(1) $\Xi_s = \sum_N \lambda_s^N Z_s^N = \qquad \lambda_s = e^{\beta\mu_s} \qquad Z_s = \sum e^{-\beta E_s} = e^{\beta\epsilon}/2\sinh\left(\frac{\beta\hbar\omega}{2}\right)$

$= \frac{1}{1-\lambda_s Z_s}$

(2) $N_s = \frac{1}{\beta}\frac{\partial}{\partial\mu_s}\ln\Xi_s = \lambda_s\frac{\partial}{\partial\lambda_s}\ln\Xi_s = \frac{\lambda_s Z_s}{1-\lambda_s Z_s} \qquad \lambda_s Z_s = \frac{N_s}{1+N_s} \rightarrow \lambda_s = \frac{1}{Z_s}$

(3) $\Xi_g = \sum_N \lambda_g^N \frac{Z_g^N}{N!} = e^{\lambda_g Z_g} \qquad \lambda_g = e^{\beta\mu_g} \qquad Z_g = \frac{V}{h^3}(2\pi m kT)^{\frac{3}{2}}$

$N_g = \lambda_g \frac{\partial}{\partial\lambda_g}\ln\Xi_g = \lambda_g Z_g \qquad \lambda_g = \frac{N_g}{Z_g}$

(4) $\lambda_s = \lambda_g \qquad N_g = \frac{Z_g}{Z_s} = \frac{V}{h^3}(2\pi m kT)^{\frac{3}{2}} e^{-\beta\epsilon} \cdot 2\sinh\left(\frac{\beta\hbar\omega}{2}\right)$

$\frac{N_g}{V} = \left(\frac{2\pi m kT}{h^2}\right)^{\frac{3}{2}} e^{-\beta\epsilon} \cdot 2\sinh\left(\frac{\hbar\omega}{2kT}\right) \approx \left(\frac{2\pi m}{h^2}\right)^{\frac{3}{2}} \hbar\omega\sqrt{kT} \qquad kT \gg \epsilon,\ kT \gg \hbar\omega$

$PV = \frac{1}{\beta}\ln\Xi_g = \frac{1}{\beta}\lambda_g Z_g = \frac{N_g}{\beta} \rightarrow P = \frac{N_g kT}{V}$

(5) $kT_c = \left(\frac{N_g}{V}\right)^2 \frac{h^4}{\omega^2}\left(\frac{m}{2\pi}\right)^3 = \left(\frac{P}{kT_c}\right)^2 \frac{h^4}{\omega^2}\left(\frac{m}{2\pi}\right)^3 \rightarrow kT_c = P^{\frac{2}{3}}\left(\frac{h^2}{\omega}\right)^{\frac{2}{3}}\frac{m}{2\pi} = \frac{m h^2}{2\pi}\left(\frac{P}{\hbar\omega}\right)^{\frac{2}{3}}$

## 44

(1) $A+B \rightleftharpoons C+Q$ $\quad \lambda_x = e^{\frac{\mu_x}{kT}}$ $\quad \Xi = \sum \lambda_A^{N_A} \lambda_B^{N_B} \lambda_c^{N_c} \frac{Z_{iA}^{N_A}}{N_A!} \frac{Z_{iB}^{N_B}}{N_B!} \frac{Z_{ic}^{N_c}}{N_c!}$

$(-\mu_A)+(-\mu_B) = (-\mu_c)+Q$

$= e^{\lambda_A Z_{iA} + \lambda_B Z_{iB} + \lambda_c Z_{ic}}$

(2) $N_A = \lambda_A \frac{\partial}{\partial \lambda_A} \ln \Xi = \lambda_A Z_{iA}$ $\quad N_B = \lambda_B Z_{iB}$ , $N_c = \lambda_c Z_{ic}$

(3) $k = \frac{N_c}{N_A N_B} = \frac{\lambda_c}{\lambda_A \lambda_B} \cdot \frac{Z_{ic}}{Z_{iA} Z_{iB}} = e^{+\frac{Q}{kT}} \frac{Z_{ic}}{Z_{iA} Z_{iB}}$

## 45

(1) 状態密度 $D(\varepsilon) = \tau \varepsilon^{\frac{1}{2}}$ 分布関数 Fermion $f(\varepsilon) = \frac{1}{e^{\beta(\varepsilon-\mu)}+1}$

$N = \int D(\varepsilon) f(\varepsilon) d\varepsilon = \tau \beta^{-\frac{3}{2}} F_{\frac{1}{2}}(\alpha) \Gamma(\frac{3}{2})$

$E = \int D(\varepsilon) \varepsilon f(\varepsilon) d\varepsilon = \tau \beta^{-\frac{5}{2}} F_{\frac{3}{2}}(\alpha) \Gamma(\frac{5}{2})$ $\quad \frac{3}{2}\Gamma(\frac{3}{2})$

$\Big\} \quad N_A \frac{E}{N} = \frac{N_A}{\beta} \frac{3}{2} \frac{F_{\frac{3}{2}}(\alpha)}{F_{\frac{1}{2}}(\alpha)}$

$\Omega = -\frac{1}{\beta} \int d\varepsilon D(\varepsilon) \ln(1+e^{\beta(\mu-\varepsilon)})$ $\quad -\frac{2}{3}E$

$= -\frac{1}{\beta} \left[ \frac{2}{3} \tau \varepsilon^{\frac{3}{2}} \ln(1+e^{\beta(\mu-\varepsilon)}) \right]_0^\infty + \frac{1}{\beta} \int d\varepsilon \frac{2}{3} \tau \varepsilon^{\frac{3}{2}} \frac{-\beta e^{\beta(\mu-\varepsilon)}}{1+e^{\beta(\mu-\varepsilon)}} = -\frac{2}{3} \tau \beta^{-\frac{5}{2}} F_{\frac{3}{2}}(\alpha) \Gamma(\frac{5}{2})$

$S = -\frac{\partial F}{\partial T} = \frac{k\alpha^2}{\mu} \frac{\partial}{\partial \alpha} (\mu N + \Omega) =$

(2) $\int D(\varepsilon)\,d\varepsilon = \frac{1}{h^3}\int dp\,dq = \frac{V}{h^3}\int 4\pi p^2 dp = \frac{V}{h^3}\frac{4\pi}{3}(2m\varepsilon)^{\frac{3}{2}}$

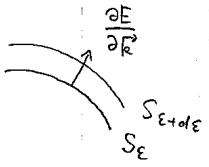
$N = \frac{V}{h^3}\frac{4\pi}{3}(2m\mu_0)^{\frac{3}{2}} \rightarrow \mu_0 = \frac{1}{2m}\left(\frac{N}{V}\frac{3h^3}{4\pi}\right)^{\frac{2}{3}} = \frac{h^2}{2m}\left(\frac{3}{4\pi}\right)^{\frac{2}{3}}\left(\frac{N}{V}\right)^{\frac{2}{3}}$

i) $\frac{h^2}{2m_e k_B}\left(\frac{3}{4\pi}\right)^{\frac{2}{3}}\left(\frac{N}{V}\right)^{\frac{2}{3}} = 2\times 10^{-10}\times\left(\frac{1}{4}\right)^{\frac{2}{3}}\times\left(10^{22}\right)^{\frac{2}{3}}\,K = 4\times 10^{4}\,K$

ii) $\frac{h^2}{2\cdot 3m_p k_B}\left(\frac{3}{4\pi}\right)^{\frac{2}{3}}\left(\frac{0.07}{3m_p}\right)^{\frac{2}{3}} = 10^{-13}\left(\frac{1}{4}\right)^{\frac{2}{3}}\left(2\times 10^{22}\right)^{\frac{2}{3}}\,K = 2\times 10^{1}\,K$

## 46

(1) $E(\vec{k}) = \varepsilon \rightarrow \frac{\partial E}{\partial \vec{k}}\cdot d\vec{k} = d\varepsilon$




$D(\varepsilon)\,d\varepsilon = 2\times V\hbar^3\int_{\varepsilon}^{\varepsilon+d\varepsilon} d\vec{k} = 2V\hbar^3\int_{S_\varepsilon} dS\cdot\frac{d\varepsilon}{\left|\frac{\partial E}{\partial \vec{k}}\right|}$

$D(\varepsilon) = 2V\hbar^3\int_{S_\varepsilon}\frac{dS}{\left|\frac{\partial E}{\partial \vec{k}}\right|}$

$N(\varepsilon) = \int_0^{\varepsilon} d\varepsilon' D(\varepsilon')$

(2) $PV = \frac{1}{\beta}\int d\varepsilon\, D(\varepsilon)\ln\left(1\pm e^{\beta(\mu-\varepsilon)}\right)^{\pm 1} =$

$= \pm\frac{1}{\beta}\left[N(\varepsilon)\ln\left(1\pm e^{\beta(\mu-\varepsilon)}\right)\right]_0^{\infty} \mp\frac{1}{\beta}\int N(\varepsilon)\,\frac{\mp\beta e^{\beta(\mu-\varepsilon)}}{1\pm e^{\beta(\mu-\varepsilon)}}\,d\varepsilon = \int N(\varepsilon)\,\frac{1}{e^{\beta(\varepsilon-\mu)}\pm 1}\,d\varepsilon$

$D(\varepsilon) = \varepsilon^l \quad n\times\mp \quad N(\varepsilon) = \frac{1}{l+1}\varepsilon^{l+1} = \frac{1}{l+1}D(\varepsilon)\varepsilon$

$PV = \frac{1}{l+1}E$

## 47

(1) $\Xi = \sum e^{\beta(N\mu - E)}$

$d(\ln\Xi) = \sum (N\mu - E)\,d\beta\,\frac{e^{\beta(N\mu-E)}}{\Xi} + \sum\beta\left(d(N\mu) - dE\right)\frac{e^{\beta(N\mu-E)}}{\Xi}$

$= N\mu\,d\beta - \langle E\rangle d\beta + \beta\,d(N\mu) - \beta\langle dE\rangle$

(2) $\Xi = \prod_\varepsilon \left( \sum_{n=0}^{\infty} e^{\beta(\mu-\varepsilon)n} \right)^{g(\varepsilon)d\varepsilon} = \prod_\varepsilon \left( \frac{1}{1-e^{\beta(\mu-\varepsilon)}} \right)^{g(\varepsilon)d\varepsilon}$

$$PV = \frac{1}{\beta} \ln \Xi = \frac{1}{\beta} \int_0^\infty d\varepsilon\, g(\varepsilon) \ln \left(1 - e^{\beta(\mu-\varepsilon)}\right)^{-1}$$

$$= \frac{1}{\beta} \left[ \underbrace{G(\varepsilon)}_{\downarrow 1} \ln \left(1 - e^{\beta(\mu-\varepsilon)}\right)^{-1} \right]_0^\infty - \frac{1}{\beta} \int_0^\infty d\varepsilon\, G(\varepsilon) \frac{-\beta e^{\beta(\mu-\varepsilon)}}{1 - e^{\beta(\mu-\varepsilon)}}$$

$$= -\frac{1}{\beta} \ln \left(1 - e^{\beta\mu}\right) + \int_0^\infty d\varepsilon\, G(\varepsilon) \frac{1}{e^{\beta(\varepsilon-\mu)} - 1}$$

$$N = N_0 + N' = \frac{1}{e^{-\beta\mu} - 1} + \int_0^\infty d\varepsilon\, g(\varepsilon) \frac{1}{e^{\beta(\varepsilon-\mu)} - 1}$$

## 48

(1) $\varepsilon_n = \varepsilon + \frac{a}{V} n^2 \qquad g(\varepsilon) = \frac{V}{a} \qquad G(\varepsilon') = \int_\varepsilon^{\varepsilon'} \frac{V}{a} d\varepsilon' = \frac{V}{a}(\varepsilon' - \varepsilon)$

$$N = N_0 + N' = \frac{1}{e^{-\beta\mu} - 1} + \int_\varepsilon^\infty d\varepsilon \frac{(V/a)}{e^{\beta(\varepsilon-\mu)} - 1} \leftarrow \frac{1}{\beta} \frac{V}{a} \int_\varepsilon^\infty d\varepsilon' \frac{\beta e^{-\beta(\varepsilon'-\mu)}}{1 - e^{-\beta(\varepsilon'-\mu)}}$$

$$= \frac{1}{\beta} \frac{V}{a} \ln \left(1 - e^{-\beta(\varepsilon'-\mu)}\right) \Big|_\varepsilon^\infty$$

$$= -\frac{1}{\beta} \frac{V}{a} \ln (1 - z)$$

$$PV = \int_\varepsilon^\infty d\varepsilon' \frac{\frac{V}{a}(\varepsilon' - \varepsilon)}{e^{\beta(\varepsilon'-\mu)} - 1} - \frac{1}{\beta} \ln \left(1 - e^{\beta\mu}\right)$$

$$\uparrow$$

$$\frac{1}{\beta^2} \frac{V}{a} \int_0^\infty dy' \frac{y'}{z^{-1} e^{y'} - 1} \qquad y' = \beta(\varepsilon' - \varepsilon)$$

(2) $p = \frac{N_0}{V} - \frac{kT}{a} \ln \left(1 - e^{\beta(\mu-\varepsilon)}\right) \qquad V \to \infty \;;\; N_0 \to \infty$

$$p_c = \frac{kT}{a} \ln \left(1 - e^{\frac{\mu-\varepsilon}{kT}}\right)^{-1}$$

(3) $p = \frac{(kT)^2}{a} \int_0^\infty dy' \underbrace{\frac{y'}{e^{-\mu\beta} e^{y'} - 1}}_{\downarrow 1} = \frac{(kT)^2}{a} \underbrace{\zeta(2)}_{\frac{\pi^2}{6}} \qquad N_0 = \frac{1}{e^{-\beta\mu} - 1} \to \infty$

## 49

(1) $g(\varepsilon) = \text{const}\, V \varepsilon^{\alpha}$

$$N_0 = \frac{1}{z^{-1}-1} \quad , \quad N' = \int_0^\infty \frac{\text{const}\, V\, \varepsilon^{\alpha}}{z^{-1} e^{\beta\varepsilon} - 1} d\varepsilon = \frac{\text{const}\cdot V}{\beta^{1+\alpha}} \underbrace{\int_0^\infty \frac{y^{\alpha}}{e^{y}-1} dy}$$

$$\Gamma(1+\alpha) = \alpha! = \alpha\Gamma(\alpha) \qquad \Gamma(1+\alpha)\,\zeta(1+\alpha)$$

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \qquad \zeta(1) = \infty$$

(2) 2次元, $\alpha = 0$ $\quad$ $l$次元 $\varepsilon(p) = a p^{l} \propto \varepsilon^{\frac{l}{2}} \propto G(\varepsilon)$ $\quad$ $g(\varepsilon) \propto \varepsilon^{\frac{l}{2}-1}$ $\quad$ $\frac{l}{2} - 1 > 0$

$l > 2$

## 50

(1) $D(\varepsilon)d\varepsilon = 2 \times \left(\frac{L}{2\pi}\right)^3 d^3k = \frac{2V}{(2\pi)^3} 4\pi k^2 dk = \frac{8\pi V}{(hc)^3} \varepsilon^2 d\varepsilon$ $\quad$ $\vec{k} = \frac{2\pi}{L}\vec{n}$

$$\Xi = \prod_{\varepsilon} \left(\frac{1}{1-e^{-\beta\varepsilon}}\right)^{D(\varepsilon)d\varepsilon} =$$

$$\frac{x^3}{3}\ln(1-e^{-x})\Big|_0^\infty - \int \frac{x^3}{3}\frac{1}{e^x-1}dx$$

$$\downarrow$$

$$PV = kT \ln \Xi = kT \int_0^\infty D(\varepsilon)d\varepsilon \ln \frac{1}{1-e^{-\beta\varepsilon}} = -\frac{8\pi V}{(hc)^3}(kT)^4 \int_0^\infty dx\, x^2 \ln(1-e^{-x})$$

$$= \frac{8\pi V}{3(hc)^3}(kT)^4 \underbrace{\int_0^\infty \frac{x^3}{e^x-1}dx}_{\pi^4/15} = \frac{4}{c}\cdot\underbrace{\frac{2\pi^5 k^4}{45 h^3 c^2}}_{\sigma} VT^4$$

$F = -pV$

$$S = -\left(\frac{\partial F}{\partial T}\right)_V = \frac{16\sigma}{3c} VT^3 \qquad U = F + TS = (-1+4)pV$$

$PV = \frac{1}{l+1}U$ $\quad$ ← 46 (2) $\quad$ $dS = 0 \rightarrow PV^{\frac{l+2}{l+1}} = \text{const}$ $\quad$ $\frac{l+2}{l+1} = \gamma$ $\quad$ $PV = (\gamma-1)U$

(2)

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$$Z = \sum_{\sigma_i} e^{\beta \sum_i \mu H \sigma_i + J \sigma_i \sigma_{i+1}} = \sum_{\sigma_i} e^{\beta \sum_i V_{i,i+1}}$$

$$= \sum_{\sigma_i} A_{12} A_{23} \cdots A_{N1} = \mathrm{tr}(A^N) = \lambda_+^N$$

$$V_{i,i+1} = \frac{1}{2}\mu H(\sigma_i + \sigma_{i+1}) + J\sigma_i\sigma_{i+1}$$

$$A = e^{\beta V_{i,i+1}} = \begin{pmatrix} e^{\beta(\mu H + J)} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta(-\mu H + J)} \end{pmatrix}$$

固有値 $\lambda_\pm = e^{\beta J}\left(\cosh \beta\mu H \pm \sqrt{\sinh^2 \beta\mu H + e^{-4\beta J}}\right)$

(1) $H = 0$;

$$\lambda_+ = e^{\beta J}(1 + e^{-2\beta J}) = 2\cosh \beta J$$

$$Z = (2\cosh \beta J)^N$$

(2) $Z$ は $\beta$ a analytic fn

$$F = -\frac{1}{\beta} \ln Z = -\frac{N}{\beta} \ln(2\cosh \beta J)$$

$$U = -\frac{\partial}{\partial \beta} \ln Z = -NJ \tanh \beta J \qquad C = \frac{\partial U}{\partial T} = Nk \left(\frac{\beta J}{\cosh \beta J}\right)^2$$

(3) $$M = \sum_i \mu \langle \sigma_i \rangle = \frac{1}{\beta} \frac{\partial}{\partial H} \ln Z = N\mu \frac{\sinh \beta\mu H \left(\sqrt{\quad} + \cosh \beta\mu H\right)}{\cosh \beta\mu H \sqrt{\quad} + \sinh^2 \beta\mu H + e^{-4\beta J}} \doteqdot N\mu^2 \cdot e^{2\beta J} \beta H$$

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1次元 ↑↑↑↓↑↑

$\Delta E = 2\mu H + 4J \sim O(1)$

$\Delta S = k \ln N \sim O(\ln N)$

$\Delta F = \Delta E - T\Delta S < 0$ 安定

2次元

↑↑↑↑
↑↓↑↑
↑↑↑↑

$\Delta E = N(2\mu H + 4J) \sim O(N)$

$\Delta S = k \ln N \sim O(\ln N)$

$\Delta F = \Delta E - T\Delta S > 0$ 不安定

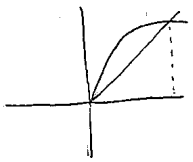
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(1) $Z = \sum_{\sigma_i=\pm 1} \cdots e^{\beta(Jz\langle\sigma\rangle\sigma_i + \mu H\sigma_i)} = \{2\cosh\beta(Jz\langle\sigma\rangle + \mu H)\}^N$

$$\beta\mu N\langle\sigma\rangle = \frac{\partial}{\partial H}\ln Z = N\tanh\beta(Jz\langle\sigma\rangle + \mu H)\,\beta\mu$$

$$\therefore \langle\sigma\rangle = \tanh\beta(Jz\langle\sigma\rangle + \mu H) \quad \cdots (*)$$

(2) $H=0$ ; $\langle\sigma\rangle = \tanh\beta Jz\langle\sigma\rangle$ $\qquad \tanh x = x - \frac{x^3}{3}$



$\langle\sigma\rangle \neq 0 \quad 1 < \beta Jz$

$1 = \frac{Jz}{kT_c}$

(3) $\langle\sigma\rangle = \frac{\beta\mu H}{1-\beta Jz}$ $\qquad M = \frac{N}{V}\mu\langle\sigma\rangle$

$$\chi = \left(\frac{\partial M}{\partial H}\right)_T = \frac{N}{V}\frac{\beta\mu^2}{1-\beta Jz} = \frac{N}{V}\frac{\mu^2}{kT-Jz} = \frac{N}{V}\frac{\mu^2}{k(T-T_c)}$$

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$$\left.\frac{\partial F}{\partial M}\right|_{T,H} = F_2(T)\,2M + F_4(T)\,4M^3 - H = 0 \quad \cdots ①$$

$$\left.\frac{\partial^2 F}{\partial M^2}\right|_{T,H} = 2F_2(T) + 12F_4(T)M^2 > 0 \quad \cdots ②$$

(1) $T > T_c$ のとき ② が $M=0$ で成り立つので $F_2 > 0$

① で $H=0$ のとき $M=0$ 以外の解がない条件は $F_4 > 0$

$T < T_c$ のとき ① で $H=0$ のとき $M=0$ 以外の解がある → $F_2 F_4 < 0$

② で $M \neq 0$ $\quad 2F_2 + 12F_4M^2 = \frac{H}{M} + 8F_4M^2 > 0 \rightarrow F_4 > 0$

(2) $U = F + TS = F - T\frac{\partial F}{\partial T} = F_0(T) - TF_0(T)' - aT_cM^2 + bM^4 \quad (H=0)$

$C = \frac{\partial U}{\partial T} = -TF_0(T)'' \quad (T > T_c,\ M=0)$

$M \neq 0 \quad \frac{\partial F}{\partial M} = a(T-T_c)2M + b\cdot 4M^3 = 0 \rightarrow M^2 = \frac{a}{2b}(T_c - T)$

$$U = F_0(T) - TF_0(T)' - \frac{a^2}{2b}T_c(T_c-T) + \frac{a^2}{4b}(T_c-T)^2$$

$$C = \left.\frac{\partial U}{\partial T}\right|_{T_c} = -TF_0(T)'' + \frac{a^2}{2b}T_c^2$$

(3) $\frac{\partial F}{\partial M} = 2a(T_c-T)M + 4bM^3 - H = 0$

$$\frac{\partial M}{\partial H} = \frac{1}{2a(T_c-T)+12bM^2} = \frac{M}{3H - 4a(T_c-T)M} \rightarrow \frac{1}{4a(T-T_c)} \quad (H \to 0)$$

$\rightarrow \infty \quad (T \to +T_c)$

$\rightarrow -\infty \quad (T \to -T_c)$

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(1) 最小仕事 $W_{min} = \Delta E - T\Delta S + P\Delta V$

(2) $\Delta E = \underset{T}{\frac{\partial E}{\partial S}}\Delta S + \underset{-P}{\frac{\partial E}{\partial V}}\Delta V + \frac{1}{2}\left(\frac{\partial^2 E}{\partial S^2}(\Delta S)^2 + 2\frac{\partial^2 E}{\partial S \partial V}(\Delta S)(\Delta V) + \frac{\partial^2 E}{\partial V^2}(\Delta V)^2\right)$

$$\Delta T = \Delta\left(\frac{\partial E}{\partial S}\right) = \frac{\partial^2 E}{\partial S^2}\Delta S + \frac{\partial^2 E}{\partial S \partial V}\Delta V$$

$$-\Delta P = \Delta\left(\frac{\partial E}{\partial V}\right) = \frac{\partial^2 E}{\partial S \partial V}\Delta S + \frac{\partial^2 E}{\partial V^2}\Delta V$$

$$= T\Delta S - P\Delta V + \frac{1}{2}\left((\Delta T)\cdot(\Delta S) - (\Delta P)\cdot(\Delta V)\right)$$

(3) $\Delta S = \frac{\partial S}{\partial T}\Delta T + \frac{\partial S}{\partial V}\Delta V = \frac{C_V}{T}\Delta T + \frac{\partial P}{\partial T}\Delta V$ $\qquad \left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$

$$\Delta P = \frac{\partial P}{\partial T}\Delta T + \frac{\partial P}{\partial V}\Delta V = \frac{\partial P}{\partial T}\Delta T - \frac{1}{\chi_T V}\Delta V$$

$$W_{min} = \frac{C_V}{T}(\Delta T)^2 + \frac{1}{\chi_T V}(\Delta V)^2$$

(4) $(\Delta T)\cdot(\Delta V) = 0$ $\qquad kT = \frac{C_V}{T}(\Delta T)^2 = \frac{1}{\chi_T V}(\Delta V)^2$

(5)

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$\overline{N_A}$

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$$\left(M\frac{d}{dt}+\frac{1}{B}\right)V = F(t) \qquad V = e^{-\frac{t}{MB}}u \quad \rightarrow \quad u = \frac{1}{M}\int_{t_0}^{t} e^{\frac{t'}{MB}} F(t')\,dt' + u(t_0)$$

$$V(t) = V(t_0)\,e^{-\frac{t-t_0}{MB}} + \frac{1}{M}\int_{t_0}^{t} e^{\frac{t'-t}{MB}} F(t')\,dt'$$

(1)